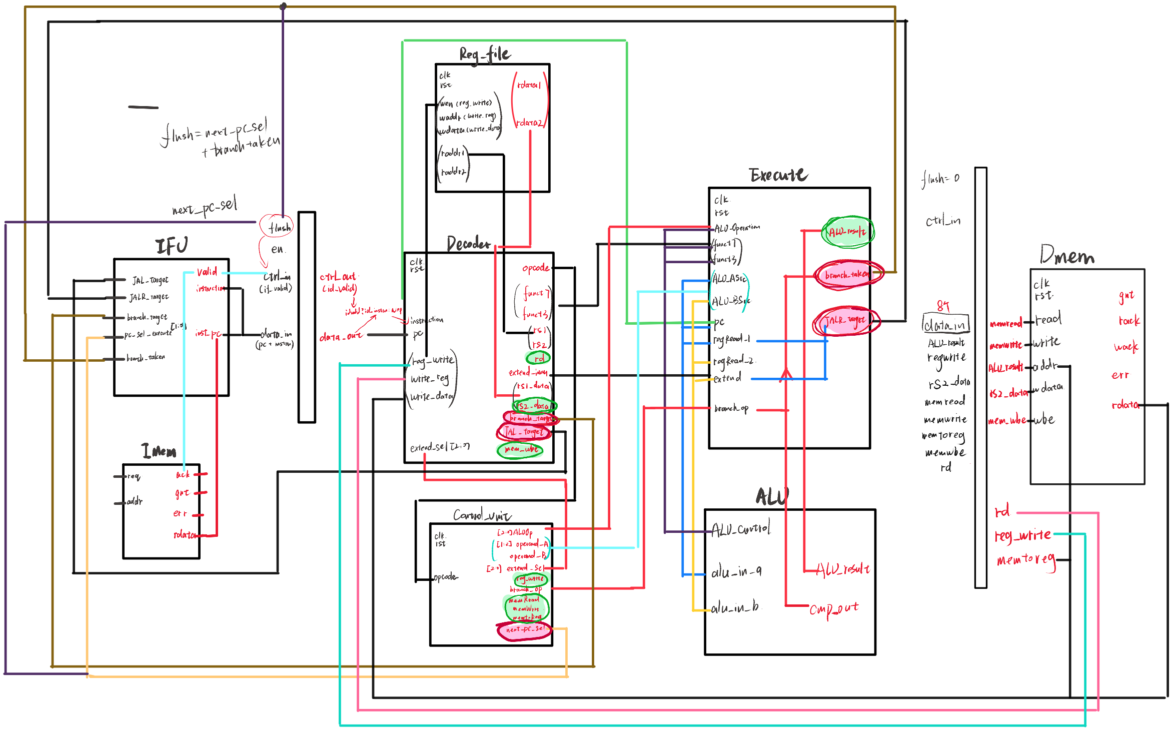
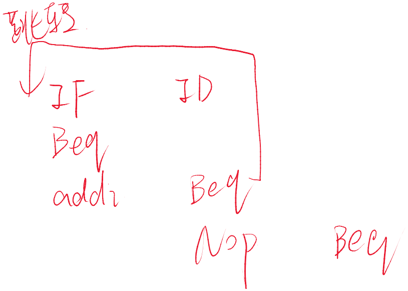
Title

Jnotes


flush=next-pc_sel
+ branch-taken
next_pc-sel
flush
en.
IFU
JAL_Target
JALR_target
branch_target
pc_sel_execute [1:0]
branch_taken
Valid
instruction
inst_pc
ctrl_in
(if_valid)
data_in
(pc + instru)
Imem
req
addr
ack
gnt
err
rdata
ctrl_out
(id_valid)
idvalid ? id_instru : Nop
data_out
Reg_file
clk
rst
wen (reg_write)
waddr (write_reg)
wdata (write_data)
raddr1
raddr2
rdata1
rdata2
Decoder
clk
rst
instruction
pc
reg_write
write_reg
write_data
extend_sel [2:0]
opcode
funct7
funct3
rs1
rs2
rd
extend_imm
rs1_data
rs2_data
branch_target
JAL_target
mem_wbe
Control_unit
clk
rst
opcode
[2:0] ALUOp
[1:0] operand_A
operand_B
[2:0] extend_Sel
reg_write
branch_op
mem Read
memWrite
memtoReg
next_pc_sel
Execute
clk
rst
ALU_Operation
funct7
funct3
ALU_ASrc
ALU_BSrc
pc
regRead_1
regRead_2
extend
branch_op
ALU_result
branch_taken
JALR_target
ALU
ALU_Control
alu_in_a
alu_in_b
ALU_result
cmp_out
flush=0
ctrl_in
84
data_in
ALU_result
regwrite
rs2_data
mem read
memwrite
memtoreg
memwbe
rd
Dmem
clk
rst
memread
read
memwrite
write
ALU_result
addr
rs2_data
wdata
mem_wbe
wbe
gnt
rack
wack
err
rdata
rd
reg_write
memtoreg



flush= pc_sel_execut + branch taken

IF	ID Ex	MEM WB
Jal		
addi	Jal	
new	addi Nop	Jal

ID Ex 检测到跳转 → 拉高 Flush=1
IF 正在顺序取指
更新目标地址

Conunit: regwrite 写使能.
memRead 内存读使能
memWrite 内存写使能
memtoReg 内存读结果写入寄存器

next_pc_sel: 跳转地址类型 [1:0]
B : 01
JAL : 10
JALR 11
else : 0

jump_flush ✗